

# WEEK-7

## MOOC EXAM NOTES – PART 1 (WATER BASICS + RUNOFF)

### Concepts Covered

- Water quantity
- Water quality

#### MCQ TRAP:

- Quantity = **how much water**
- Quality = **how clean/safe water**

### Hydrologic Cycle (VERY IMPORTANT)

#### Key Processes:

- **Evaporation** → water → vapor
- **Transpiration** → plants release water
- **Condensation** → cloud formation
- **Precipitation** → rain/snow
- **Runoff** → flow on surface
- **Infiltration** → water into soil

#### IMPORTANT FACT:

- Units of water transfer =  **$10^{12} \text{ m}^3/\text{year}$**

#### MCQ TRAP:

- Transpiration  $\neq$  evaporation
- Runoff = **surface flow**, not underground

### Sources of Water

#### 1. Groundwater

- Located in **aquifers**
- Extracted using **deep wells**

## 2. Surface Water

- Found in:
  - **Lakes**
  - **Rivers**
  - **Streams**
- Used via **intake system**

👉 MCQ TRAP:

- Intake is used for **surface water**, NOT groundwater

## Aquifer Types (HIGHLY IMPORTANT)

Types:

- **Unconfined Aquifer**
  - Open to surface
  - Has **water table**
- **Confined Aquifer**
  - Between **confining beds**
  - Under pressure
- **Artesian Well**
  - Water rises **automatically**

👉 MCQ TRAP:

- Flowing artesian well = **no pumping needed**

## Water Supply System (FLOW PROCESS)

Sequence:

1. **Reservoir water**
2. **Intake**
3. **Treatment plant**
4. **Coagulation & Flocculation**
5. **Sedimentation**
6. **Filtration**

7. **Disinfection**
8. **Storage & distribution**

👉 MCQ TRAP:

- Disinfection happens **after filtration**

## **Essential Elements of Water Supply**

1. **Intake & reservoir**
2. **Treatment plant**
3. **Elevated tanks & standpipes**
4. **Valves**
5. **Hydrants**
6. **Distribution system**
7. **Service connections**

👉 MCQ TIP:

- Order questions are common

## **Watershed (VERY IMPORTANT DEFINITION)**

**Definition:**

- Area where **all water drains to same outlet (river)**

**Example:**

- Columbia River Basin → Pacific Ocean

👉 MCQ TRAP:

- Watershed = **drainage area**, NOT just river

## **Runoff Estimation (MOST IMPORTANT NUMERICAL)**

 **Formula:**

$$Q = CIA360 \quad Q = 360CIA$$

**Where:**

- **Q** = runoff ( $\text{m}^3/\text{s}$ )
- **C** = runoff coefficient
- **I** = rainfall intensity ( $\text{mm/hr}$ )
- **A** = area (hectares)

👉 MCQ TRAP:

- Divide by **360 (very important)**

### Runoff Coefficient Table (MEMORIZE)

| Area Type             | C Value        |
|-----------------------|----------------|
| <b>Business areas</b> | 0.70 –<br>0.90 |
| <b>Apartments</b>     | 0.50 –<br>0.70 |
| <b>Single family</b>  | 0.30 –<br>0.50 |
| <b>Parks/Lawns</b>    | 0.10 –<br>0.25 |
| <b>Paved streets</b>  | 0.80 –<br>0.90 |
| <b>Tight roofs</b>    | 0.70 –<br>0.95 |

👉 MCQ TRAP:

- Highest C → **roofs & paved streets**
- Lowest C → **parks**

### Numerical Example (IMPORTANT)

**Given:**

- Area = **300 hectares**
- Intensity = **30 mm/hr**
- Final coefficient = **0.44**

**Answer:**

$$Q=11 \text{ m}^3/\text{sec} \quad Q=11 \text{ m}^3/\text{sec}$$

👉 MCQ TRAP:

- Weighted average formula used for C:

$$C = \frac{A_1 C_1 + A_2 C_2 + \dots A_1 + A_2 + \dots}{C = \frac{A_1 + A_2 + \dots A_1 C_1 + A_2 C_2 + \dots}{A_1 + A_2 + \dots}}$$

## EXPECTED MCQs

1.

Water in aquifers is called:

- A) Surface water
- B) Groundwater ☒
- C) Rainwater
- D) Runoff

2.

Which comes first?

- A) Filtration
- B) Sedimentation ☒
- C) Disinfection
- D) Storage

3.

Highest runoff coefficient?

- A) Parks
- B) Lawns
- C) Roofs ☒
- D) Soil

4.

Unit of hydrologic cycle transfer:

- A)  $\text{m}^3/\text{day}$
- B)  $\text{m}^3/\text{year}$
- C)  $10^{12} \text{ m}^3/\text{year}$  ☒
- D) liters/sec

5.

Artesian well water:

- A) Needs pump
- B) Flows naturally 
- C) Stored in tank
- D) Surface water

### **ULTRA-IMPORTANT REVISION POINTS**

- **$Q = CIA / 360$**
- **Watershed = drainage area**
- **Aquifer types**
- **Treatment sequence**
- **Runoff coefficient values**

# MOOC EXAM NOTES – PART 2 (WATER AVAILABILITY & DEMAND)

## Water Availability (GLOBAL FACTS)

### KEY FACT:

- Total freshwater on Earth  $\approx 3.5 \times 10^7 \text{ km}^3$

### Distribution of Freshwater (VERY IMPORTANT TABLE)

| Source           | % of World Freshwater |
|------------------|-----------------------|
| Glaciers & snow  | 68.7%                 |
| Groundwater      | 30.1%                 |
| Lakes            | 0.26%                 |
| Soil moisture    | 0.05%                 |
| Atmosphere       | 0.04%                 |
| Marshes/swamps   | 0.03%                 |
| Rivers           | 0.006%                |
| Biological water | 0.003%                |

### MCQ TRAPS:

- Largest = Glaciers (NOT groundwater)
- Usable = mostly groundwater
- Rivers = extremely small (0.006%)

## Water Usage (GLOBAL)

### World Distribution:

- **Agriculture** → ~70%
- **Industry** → ~22%
- **Domestic** → ~8%

### MCQ TRAP:

- Agriculture ALWAYS highest

### Water Use by Income Groups

- **World avg:** Agriculture dominates
- **Low & middle income:** Agriculture ~81%
- **High income countries:**
  - Industry ↑
  - Agriculture ↓ (~30%)

### MCQ TRAP:

- Rich countries → **industrial use higher**

## Municipal Water Demand

### Types:

- **Domestic use**
  - Indoor → Kitchen, Laundry, Bath, Toilet
  - Outdoor → Gardening, washing
- **Non-domestic use**

### MCQ TRAP:

- Indoor vs outdoor often asked

## Types of Water Demand (VERY IMPORTANT)

1. **Domestic**
2. **Industrial**
3. **Institutional & Commercial**



4. **Public uses**
5. **Fire demand**
6. **Losses (wastage & thefts)**

👉 MCQ TRAP:

- Losses ARE considered demand



## **Per Capita Demand (PCD) FORMULA**

$PCD = \frac{\text{Total yearly water requirement}}{\text{Design population}}$   
 $PCD = \frac{365 \times \text{Design population}}{\text{Total yearly water requirement}}$

👉 Units:

- **Liters per person per day (LPCD)**

👉 MCQ TRAP:

- 365 must be included (students forget)



## **Factors Affecting Per Capita Demand**

🔥 **Must Remember All (very important):**

1. **Size of city**
2. **Climate**
3. **Habits of people**
4. **Industrial activity**
5. **Water quality**
6. **Distribution pressure**
7. **Sewerage system**
8. **Supply system**
9. **Cost of water**
10. **Metering policy**

👉 MCQ TRAP:

- Climate ↑ → water demand ↑

## Factors Affecting Water Demand (CONCEPTUAL)

| Factor     | More Water     | Less Water    |
|------------|----------------|---------------|
| Climate    | Hot            | Cold          |
| Habits     | Rich lifestyle | Simple        |
| Sewer      | Available      | Not available |
| Water rate | Cheap          | Expensive     |
| Supply     | Continuous     | Restricted    |

👉 MCQ LOGIC:

- Better facilities → **more usage**

## Variation in Water Demand

📌 Key Patterns:

- Peak → **morning & evening**
- Low → night

## Important FORMULAS (VERY IMPORTANT)

### 1. Maximum Daily Demand:

Max daily demand =  $1.8 \times \text{Average daily demand}$

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### 2. Maximum Hourly Demand:

Max hourly demand =  $2.7 \times \text{Average daily demand}$

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### 3. Total Demand:

- Max daily demand + Fire demand
- OR
- Max hourly demand

👉 MCQ TRAP:

- $2.7 = 1.5 \times 1.8$  (they may ask derivation)

### 🔥 Fire Demand

📌 Concept:

- Water needed for **firefighting**
- **Total yearly volume small**
- But **flow rate VERY HIGH**

👉 MCQ TRAP:

- Fire demand = **high rate, low total volume**

### 📐 Fire Demand FORMULAS (VERY IMPORTANT)

#### 1. Kuichling's Formula:

$$Q = 3182 P^{1/4} \quad Q = 3182 P^{0.25}$$

#### 2. Boston's Formula:

$$Q = 5663 P^{1/4} \quad Q = 5663 P^{0.25}$$

#### 3. Freeman's Formula:

$$Q = 1136 (P^{1/4} + 10) \quad Q = 1136 (P^{0.25} + 10)$$

#### 4. National Board Formula:

$$Q = 4637 P^{1/4} (1 - 0.01 P^{1/4}) \quad Q = 4637 P^{0.25} (1 - 0.01 P^{0.25})$$

👉 Units:

- **Q in liters/min**
- **P = population**

👉 MCQ TRAP:

- Population always in **thousands (P/1000)**

## 🎯 EXPECTED MCQs (HIGH PROBABILITY)

1.

Largest freshwater source?

- A) Groundwater
- B) Lakes
- C) Glaciers ✓
- D) Rivers

2.

Maximum hourly demand equals:

- A)  $1.5 \times \text{avg}$
- B)  $1.8 \times \text{avg}$
- C)  $2.7 \times \text{avg}$  ✓
- D)  $3.0 \times \text{avg}$

3.

Water demand increases with:

- A) High cost
- B) Cold climate
- C) Better supply ✓
- D) Less pressure

4.

Fire demand is:

- A) Low rate, high volume
- B) High rate, low volume ✓
- C) Constant
- D) Zero

5.

Unit of PCD:

- A)  $\text{m}^3/\text{sec}$

- B) L/day  
C) L/person/day   
D) m<sup>3</sup>/year

## **ULTRA-IMPORTANT REVISION LIST**

- **Freshwater distribution (68.7%, 30.1%)**
- **Agriculture = 70%**
- **PCD formula**
- **1.8 & 2.7 factors**
- **Fire demand formulas**
- **Factors affecting demand (all 10)**

# MOOC NOTES – PART 3 (POPULATION FORECASTING + CONVEYANCE)

## Population Forecasting (CORE CONCEPT)

### Definition:

- Used to estimate future population for design period
- Used in:
  - Water supply
  - Sanitation systems

### IMPORTANT:

- Underestimate → system inadequate ✖
- Overestimate → costly ✖
- Design is based on population at END of design period

## Factors Affecting Population Change

- Births → increase
- Deaths → decrease
- Migration → increase/decrease
- Annexation → increase

### MCQ TRAP:

- Migration can be both + and –

## Population Forecasting Methods (VERY IMPORTANT LIST)

1. Arithmetic Increase Method
2. Geometric Increase Method
3. Incremental Increase Method
4. Graphical Method
5. Comparative Graphical Method
6. Master Plan Method
7. Logistic Curve Method

### MCQ TIP:

- They LOVE asking “which method is suitable for \_\_\_ city”

## 1. Arithmetic Increase Method

### Concept:

- Population increases at constant rate

Formula:

$$P_n = P + nC$$

Where:

- P = present population
- C = average increase
- n = number of decades

### Use:

- Large & old cities

### MCQ TRAP:

- Gives lower estimate for new cities

## 2. Geometric Increase Method

### Concept:

- Growth is percentage-based (exponential)

Formula:

$$P_n = P(1 + I_g/100)^n$$

Where:

- I<sub>g</sub> = geometric mean growth rate

### Use:

- New industrial cities

👉 MCQ TRAP:

- Gives higher values



### 3. Incremental Increase Method



Concept:

- Combination of arithmetic + incremental growth

Formula:

$$P_n = P + nX + \frac{n(n+1)}{2}Y \quad P_n = P + nX + 2n(n+1)Y$$

Where:

- X = average increase
- Y = incremental increase



Use:

- Average cities with increasing growth



### IMPORTANT NUMERICAL VALUES (FROM YOUR EXAMPLE)

Arithmetic:

- Avg increase = 345463
- 2021 → 2931325
- 2031 → 3276788
- 2041 → 3622251

Geometric:

- Growth rate = 23.5%
- 2021 → 3193540
- 2031 → 3944021
- 2041 → 4870866

Incremental:

- Avg increase = 345463



- Incremental increase = 87729
- 2021 → 3019054
- 2031 → 3539975
- 2041 → 4148625



## Graphical Method



### Concept:

- Plot past data → extend curve



### MCQ TRAP:

- Needs experience & judgment



## Comparative Graphical Method



### Concept:

- Compare with similar cities



### Advantage:

- Works even if past data incomplete



## Master Plan Method



### Concept:

- Based on planned city zones

### Key Points:

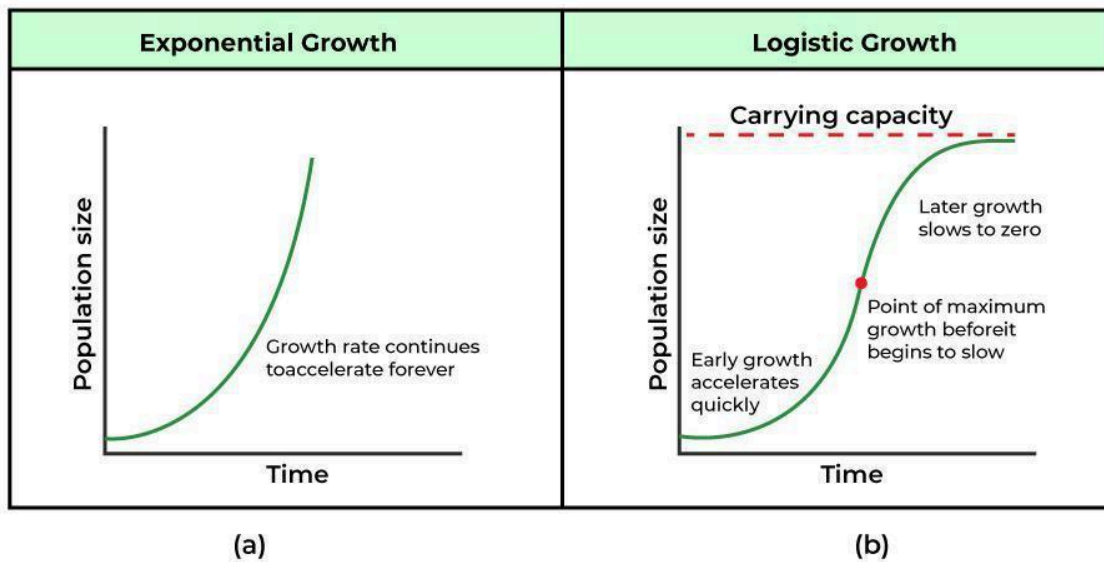
- City divided into:
  - Residential
  - Commercial
  - Industrial
- Population density predefined



### MCQ TRAP:

- Used for big metropolitan cities

## Logistic Curve Method (ADVANCED)



Formula:

$$P = \frac{P_s}{1 + e^{-nt}}$$

Where:

- $P_s$  = saturation population
- Growth is S-shaped

👉 MCQ TRAP:

- Has inflection point

## Conveyance of Water (PIPES)

 Methods:

- Pressure conduits (pipes)

Equations used:

1. Darcy-Weisbach equation
2. Hazen-Williams equation



## Darcy-Weisbach Equation

$$h_f = f \frac{LV^2}{2gd} \quad h_f = 2gdf \frac{LV^2}{2}$$



MCQ TRAP:

- Used for head loss



## Hazen-Williams Equation

$$V = 0.85 C R^{0.63} S^{0.54} \quad V = 0.85 C R^{0.63} S^{0.54}$$



MCQ TRAP:

- Used in water supply pipes



## Pump Power Formula

$$BHP = \frac{\gamma_w Q h}{\eta_p \eta_m} \quad BHP = \frac{\eta_p \eta_m \gamma_w Q h}{0.746}$$



MCQ TRAP:

- Includes efficiency of pump & motor



## Numerical Example (VERY IMPORTANT)

Given:

- Population = 1 lakh
- Supply = 200 LPCD
- Distance = 8 km
- Friction factor = 0.003
- Head = 10 m

Result:

- $Q = 0.231 \text{ m}^3/\text{s}$
- Diameter = 0.4 m



MCQ TRAP:

- Convert LPCD  $\rightarrow$   $\text{m}^3/\text{sec}$  carefully

## EXPECTED MCQs

1.

Best method for old cities:

- A) Geometric
- B) Arithmetic ☒
- C) Logistic
- D) Graphical

2.

Which gives highest population?

- A) Arithmetic
- B) Incremental
- C) Geometric ☒
- D) Graphical

3.

Logistic curve is:

- A) Linear
- B) Exponential
- C) S-shaped ☒
- D) Constant

4.

Hazen-Williams is used for:

- A) Population
- B) Pipe flow ☒
- C) Rainfall
- D) Demand

5.

Incremental method includes:

- A) Only average
- B) Only percentage
- C) Avg + incremental ☒
- D) None

## **ULTRA-IMPORTANT REVISION**

- $P_n = P + nC$
- $P_n = P(1 + I_g/100)^n$
- Incremental formula
- Logistic = S-curve
- Darcy & Hazen formulas
- Arithmetic → old city
- Geometric → new city



## MOOC NOTES – PART 4 (WATER QUALITY)



### Types of Water (BASIC CLASSIFICATION)

Types:

- Pure water
- Potable water → safe to drink ✓
- Palatable water → good taste/smell
- Wholesome water → safe + acceptable
- Polluted water
- Contaminated water
- Mineral water



MCQ TRAP:

- Potable = safe
- Palatable = taste only



### Safe / Wholesome Water Characteristics

MUST satisfy:

- Free from bacteria
- Chemically safe
- No radioactive elements
- No harmful organic substances
- Aesthetically acceptable (color, taste, smell)



MCQ TRAP:

- Safe ≠ only chemical → must include **biological + aesthetic**



### Polluted Water (DEFINITION)

- Presence of **undesirable substances** harmful to:
  - Humans
  - Vegetation
  - Property



MCQ:

- Pollution ≠ only biological → includes chemical too

## **Water Analysis – Key Factors**

- **Sampling**
- **Proper labeling**
- **Selection of parameters**
- **Accuracy & precision of method**
- **Preservation**
- **Reporting values**

👉 MCQ TRAP:

- Sampling + preservation often asked

## **Water Quality Parameters (VERY IMPORTANT)**

**Categories:**

1. **Physical**
2. **Chemical (Inorganic/Organic)**
3. **Biological**
4. **Bacteriological**
5. **Radioactive**

 **Important Parameters:**

**Physical:**

- Temperature
- **pH**
- Colour
- Taste & Odour
- Turbidity
- Conductivity
- Total dissolved solids

**Chemical:**

- Hardness
- Chlorides
- Nitrates

- Fluoride
- Metals (Fe, Cu, Zn, Pb, Hg etc.)
- Dissolved gases

#### **Biological:**

- **Bacteria**
- Protozoa
- Viruses
- Worms
- Fungi

#### **Bacteriological:**

- **Total coliform**
- **Fecal coliform**

#### **Organic:**

- **BOD**
- **COD**
- Oil & grease
- Pesticides

#### **Radioactive:**

- **Alpha emitters**
- **Beta emitters**

#### **MCQ TRAP:**

- BOD/COD = **organic pollution**
- Coliform = **bacteria indicator**

### **Characteristics of Water (3 MAIN TYPES)**

#### **1. Physical:**

- Turbidity
- Colour
- Taste & odour
- Temperature
- Conductivity



## 2. Chemical:

- pH
- Hardness
- Chloride
- Nitrogen
- Metals
- Dissolved gases

## 3. Biological:

- Microorganisms (bacteria, virus etc.)



## Water Quality Considerations

- **Irrigation** → pH, salts, nutrients
- **Domestic** → as per **BIS standards**
- **Water bodies** → as per **CPCB guidelines**
- **Industrial** → depends on process



### MCQ TRAP:

- BIS → drinking water
- CPCB → environment



## TURBIDITY (VERY IMPORTANT)



### Definition:

- Measure of **cloudiness of water**

### Cause:

- **Suspended & colloidal particles**

### Nature:

- **Optical property (light scattering)**



## Measurement Methods:

- **Secchi disk** (for lakes)

- **Nephelometer** (lab instrument)

👉 MCQ TRAP:

- Turbidity = **light scattering**, not absorption

## **COLOUR (IMPORTANT)**

📌 **Key Points:**

- Not harmful but **affects appearance**
- Can stain clothes

**Measurement:**

- Instrument → **Tintometer**
- Unit → **TCU (True Colour Unit)**

👉 MCQ TRAP:

- Colour unit = **TCU**

## **TASTE & ODOUR**

**Cause:**

- **Organic matter**
- **Inorganic salts**
- **Dissolved gases**

## **Threshold Odour Number (TON):**

- Defined as **dilution factor where odour just detectable**

**Formula:**

Dilution factor =  $\frac{V_{\text{sample}} + V_{\text{dilution}}}{V_{\text{sample}}}$  Dilution factor =  $\frac{V_{\text{sample}} + V_{\text{dilution}}}{V_{\text{sample}}}$

**Instrument:**

- **Osmoscope**

**Permissible limit:**

- 1 to 3

👉 MCQ TRAP:

- TON is based on **dilution**, not concentration

## 🎯 EXPECTED MCQs (VERY HIGH PROBABILITY)

1.

Safe drinking water is called:

- A) Pure
- B) Palatable
- C) Potable ✓
- D) Mineral

2.

Turbidity is due to:

- A) Dissolved salts
- B) Suspended particles ✓
- C) Temperature
- D) pH

3.

Instrument for colour:

- A) Nephelometer
- B) Tintometer ✓
- C) Osmoscope
- D) pH meter

4.

Coliform indicates:

- A) Chemical pollution
- B) Bacterial contamination ✓
- C) Temperature
- D) Hardness

5.

Unit of colour:

- A) ppm
- B) NTU
- C) TCU ☒
- D) mg/L

6.

Threshold odour number depends on:

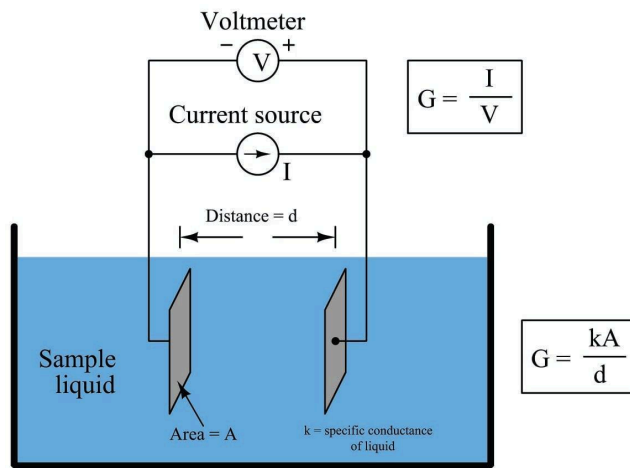
- A) Temperature
- B) Dilution factor ☒
- C) Pressure
- D) pH

### **ULTRA-IMPORTANT FINAL REVISION**

- Potable vs Palatable
- Turbidity = light scattering
- Tintometer → colour
- Nephelometer → turbidity
- Coliform → bacteria
- BOD/COD → organic pollution
- TON → dilution factor
- TCU unit

## **MOOC NOTES – PART 5 (ADVANCED WATER QUALITY + NUMERICALS)**

## ⚡ Conductivity (VERY IMPORTANT)



### 📌 Definition:

- Ability of water to conduct electric current

### 📏 Formula:

$$K = LA \times G \quad K = AL \times G$$

### Where:

- $G$  = conductance
- $L$  = distance
- $A$  = area

### 📏 Units:

- $\mu\text{S}/\text{cm}$  or  $\mu\text{mho}/\text{cm}$

### 📌 Depends on:

- Ion concentration
- Mobility
- Valency
- Temperature

### 📌 Uses:

- Check purity of water

- Estimate TDS
- Monitor seasonal variation

👉 MCQ TRAP:

- Higher ions → higher conductivity

## 🔔 TON NUMERICAL (IMPORTANT)

Given:

- Sample = 10 ml
- Distilled water = 190 ml

Formula:

$$TON = \frac{A+B}{A} \quad TON = \frac{A+B}{A}$$

👉 Answer:

$$TON = \frac{20}{10} \quad TON = 2$$

👉 MCQ TRAP:

- TON always based on dilution ratio



## Temperature

- Affects:
  - Chemical reactions
  - Dissolved oxygen
  - Biological activity

👉 MCQ:

- High temp → low DO (dissolved oxygen)

## ⚠️ Impurities in Water (CLASSIFICATION)

1. Suspended:

- Visible particles
- Cause turbidity

## 2. Colloidal:

- Very small particles
- Do not settle easily

## 3. Dissolved:

- Organic + inorganic + gases
- Includes:
  - Salts
  - Metals (Pb, As)

### 👉 MCQ TRAP:

- Turbidity → suspended, not dissolved



## SOLIDS (VERY IMPORTANT NUMERICAL)



### Total Solids (TS)

#### Formula:

$$TS = \frac{W_2 - W_1}{V} \times 10^6 \quad TS = \frac{W_2 - W_1}{V} \times 10^6$$

#### Where:

- W<sub>2</sub> = final weight
- W<sub>1</sub> = initial weight
- V = volume (ml)

#### Example:

- TS = 340 mg/L

### 👉 MCQ TRAP:

- Multiply by 10<sup>6</sup>



### Total Dissolved Solids (TDS)

- After filtration
- Only dissolved part

## Volatile Solids

- Burnable (organic)

### MCQ:

- Volatile solids = organic matter

## pH (VERY IMPORTANT)

Formula:

$$pH = -\log[H^+], pH + pOH = 14 \quad pH = -\log[H^+], pH + pOH = 14$$

### Key Points:

- $pH < 7 \rightarrow$  acidic
- $pH > 7 \rightarrow$  alkaline

Example:

- $pH = 9 \rightarrow [H^+] = 10^{-9}$

Another:

- $[H^+] = 10^{-6} \rightarrow pH = 6 \rightarrow pOH = 8$

### Measurement:

- Instrument  $\rightarrow$  pH meter

## Indicators:

| Indicator     | Range   | Color Change             |
|---------------|---------|--------------------------|
| Methyl orange | 2.8–4.4 | Red $\rightarrow$ Yellow |



|            |         |                 |
|------------|---------|-----------------|
| Methyl red | 4.4–6.2 | Red →<br>Yellow |
|------------|---------|-----------------|

|            |         |                 |
|------------|---------|-----------------|
| Phenol red | 6.8–8.4 | Yellow →<br>Red |
|------------|---------|-----------------|

|                     |              |                 |
|---------------------|--------------|-----------------|
| Phenolphthal<br>ein | 8.6–10.<br>3 | Yellow →<br>Red |
|---------------------|--------------|-----------------|

👉 MCQ TRAP:

- $\text{pH} + \text{pOH} = 14$  always

## 💎 HARDNESS (VERY VERY IMPORTANT)

📌 Definition:

- Due to  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  salts

🔥 Types:

### 1. Temporary Hardness

- Due to:
  - Bicarbonates
- Removed by:
  - Boiling

### 2. Permanent Hardness

- Due to:
  - Sulphates, chlorides, nitrates
- Not removed by boiling

👉 MCQ TRAP:

- Temporary = carbonate hardness

## Hardness Relations (VERY IMPORTANT)

Key formulas:

$$TH = CH + NCH \quad TH = CH + NCH$$

Definitions:

- CH (carbonate hardness) = min(TH, Alkalinity)
- NCH = TH - CH

Conditions:

- If TH > alkalinity → CH = alkalinity
- If TH ≤ alkalinity → CH = TH, NCH = 0

👉 MCQ TRAP:

- CH depends on minimum

## Hardness Numerical (IMPORTANT)

Example 1:

- TH = 550 mg/L
- CH = 200 mg/L
- NCH = 350 mg/L

Example 2:

- TH = 362.5 mg/L
- Alkalinity = 203.3 mg/L
- CH = 203.3
- NCH = 159.2

## Hardness Measurement

- Method → EDTA titration

## Chloride Content

- Present as NaCl
- Gives salty taste

👉 MCQ:

- High chloride → salty water

## 🌱 Nitrogen Content (VERY IMPORTANT)

Types:

1. Free ammonia
  - Initial decomposition
  - Limit: 0.15 mg/L
2. Organic nitrogen
  - Before decomposition
  - Limit: 0.3 mg/L
3. Nitrites
  - Partial oxidation
4. Nitrates
  - Fully oxidized
  - Indicates old pollution

👉 MCQ TRAP:

- Nitrate = old pollution
- Ammonia = fresh pollution

## 🎯 EXPECTED MCQs (FINAL HIGH PROBABILITY)

1.

Conductivity increases with:

- A) Less ions
- B) More ions ☒
- C) Temperature decrease
- D) None

2.

Temporary hardness is removed by:

- A) Filtration
- B) Boiling ☒
- C) Chlorination
- D) Sedimentation

3.

$\text{pH} + \text{pOH} =$

- A) 7
- B) 10
- C) 14 ☒
- D) 1

4.

Nitrate indicates:

- A) Fresh pollution
- B) Old pollution ☒
- C) No pollution
- D) Acid

5.

Unit of conductivity:

- A) mg/L
- B)  $\mu\text{S}/\text{cm}$  ☒
- C) NTU
- D) TCU

 **ULTIMATE REVISION (JUST BEFORE EXAM)**

**MUST REMEMBER:**

- pH formulas
- Hardness relations
- TS formula
- TON formula
- Conductivity units
- Temporary vs permanent hardness
- Nitrogen types

- Coliform, BOD, COD

 **MOOC NOTES – PART 6 (FINAL: CHEMICAL + BIOLOGICAL + HEALTH)**

## Nitrogen Content (VERY IMPORTANT)

### Key Points:

- Nitrites → highly dangerous → should be NIL ⚠
- Ammonia + Organic nitrogen = Kjeldahl nitrogen
- Nitrates ( $\text{NO}_3^-$ ):
  - Excess → causes Methemoglobinemia (Blue baby disease)
  - Limit = 45 mg/L

### MCQ TRAPS:

- Nitrite = most dangerous
- Nitrate = limit 45 mg/L

## Metals & Chemical Substances

### Limits & Effects:

| Substance | Limit    | Effect          |
|-----------|----------|-----------------|
| Iron      | 0.3 ppm  | Stains clothes  |
| Manganese | 0.05 ppm | Affects taste   |
| Copper    | 1.3 ppm  | Toxic in excess |
| Sulphate  | 250 ppm  | Laxative effect |

## Sulphate:

- Found in rainwater
- High concentration → laxative effect (with Ca & Mg)

### MCQ TRAP:

- Sulphate → digestive effect

## Fluoride (VERY IMPORTANT)

- Optimum: 0.8 – 1.0 ppm
- Low → dental caries
- High (>1.5 ppm) → fluorosis (teeth discoloration)

### MCQ TRAP:

- Too low AND too high both harmful

## Atomic Absorption Spectroscopy (AAS)

### Use:

- Detect heavy metals in water

### MCQ:

- AAS → metal detection

## Dissolved Oxygen (DO)

### Concept:

- Oxygen enters water from atmosphere
- Consumed by organic matter

### IMPORTANT:

- Low DO → organic pollution present

### Measurement:

- Winkler method

👉 MCQ TRAP:

- DO low = polluted water

## Biological Characteristics

Types of organisms:

- Bacteria
- Protozoa
- Viruses
- Worms
- Fungi

## Bacteria (VERY IMPORTANT)

📌 Features:

- Single-cell
- Size: 1–4 microns
- Reproduce by binary fission

Types:

- Non-pathogenic → harmless
- Pathogenic → disease causing

## Other Organisms

- Protozoa → eat bacteria
- Viruses → cause diseases
- Worms → larvae
- Fungi → grow without sunlight

## Bacteria Classification (OXYGEN BASED)

- Aerobic → need oxygen
- Anaerobic → no oxygen



- Facultative → both

👉 MCQ TRAP:

- Facultative = most flexible

## Pathogenic Bacteria Testing

📌 Key Point:

- Direct testing difficult
- Use coliform bacteria as indicator

👉 MCQ:

- Coliform = indicator of contamination

## Coliform Testing Methods

1. Membrane filter technique
2. MPN (Most Probable Number)

📌 Process:

- Lactose fermentation → gas formation → coliform present

## Coliform Index (VERY IMPORTANT)

📌 Definition:

- Reciprocal of smallest sample giving positive result

 Standard:

- $\leq 1$  coliform per 100 ml → safe

Formula:

$MPN = 100 \times \text{No. of positive tubes} \times (\text{ml in positive}) \times (\text{ml in negative})$   
 $MPN = (\text{ml in positive}) \times (\text{ml in negative}) \times 100 \times \text{No. of positive tubes}$

👉 MCQ TRAP:

- Unit → per 100 ml

## Water-Borne Diseases

Common Diseases:

- Cholera
- Typhoid
- Diarrhea
- Hepatitis
- Giardia

## Prevention (IMPORTANT)

- Drink filtered/boiled water
- Avoid roadside drinks
- Store water cleanly
- Wash hands properly

## FINAL EXPECTED MCQs (VERY HIGH PROBABILITY)

1.

Nitrite concentration should be:

- A) High
- B) Moderate
- C) Zero ☒
- D) 45 mg/L

2.

Disease due to nitrate:

- A) Cholera
- B) Fluorosis

C) Methemoglobinemia 

D) Typhoid

3.

Fluoride excess causes:

A) Caries

B) Fluorosis 

C) Cholera

D) Hardness

4.

DO low indicates:

A) Clean water

B) Organic pollution 

C) Hard water

D) Salty water

5.

Coliform indicates:

A) Metals

B) Temperature

C) Bacterial contamination 

D) Hardness

6.

Safe coliform level:

A) 10/100 ml

B) 5/100 ml

C)  $\leq 1/100$  ml 

D) 0



**ULTIMATE LAST-DAY REVISION (READ THIS 2 TIMES)**



**MUST REMEMBER:**

- Nitrite = dangerous (0)

- Nitrate = 45 mg/L
- Fluoride = 0.8–1.5 ppm
- DO ↓ → pollution
- Coliform → contamination
- MPN method
- Hardness formulas
- pH relations
- Conductivity units